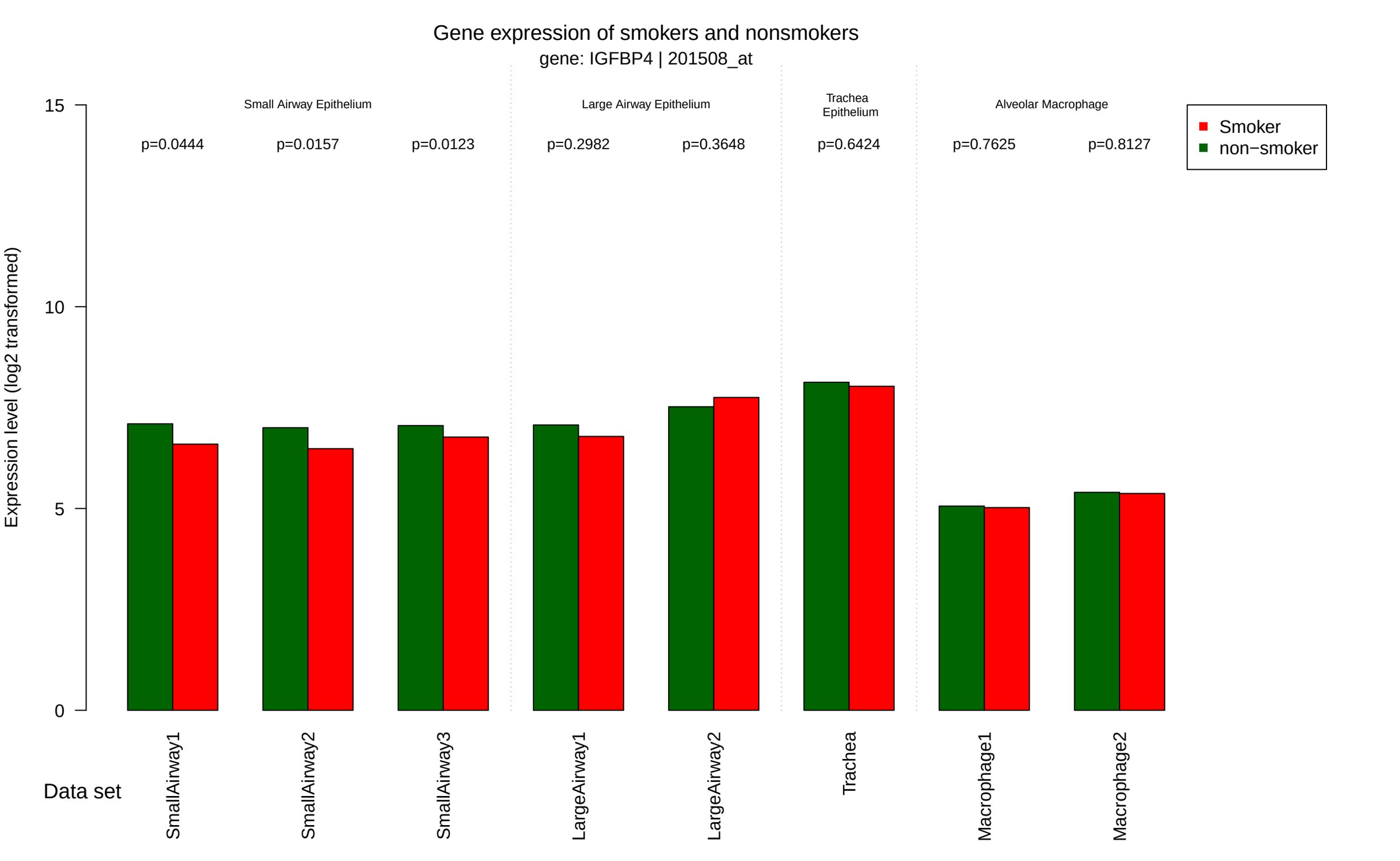
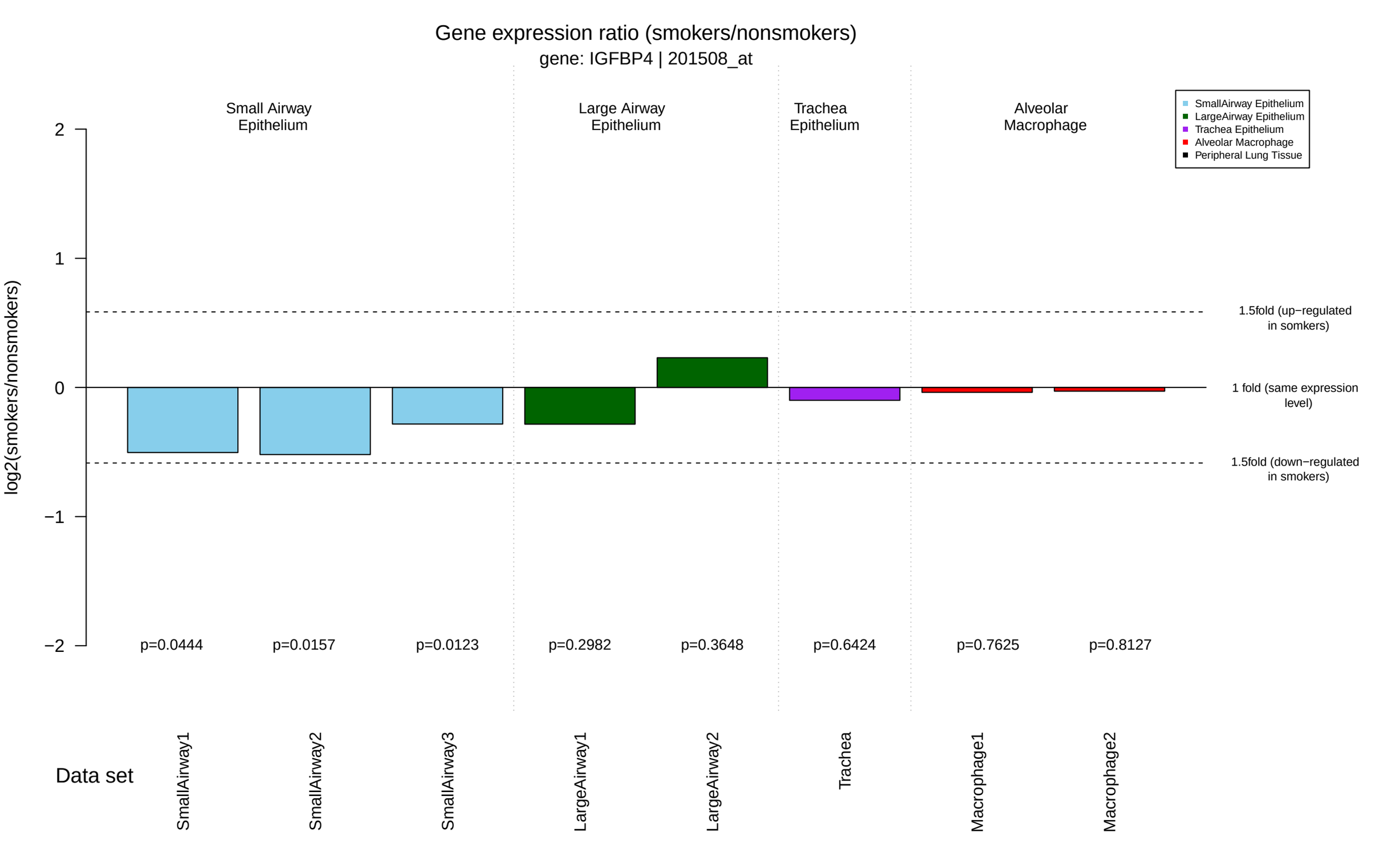
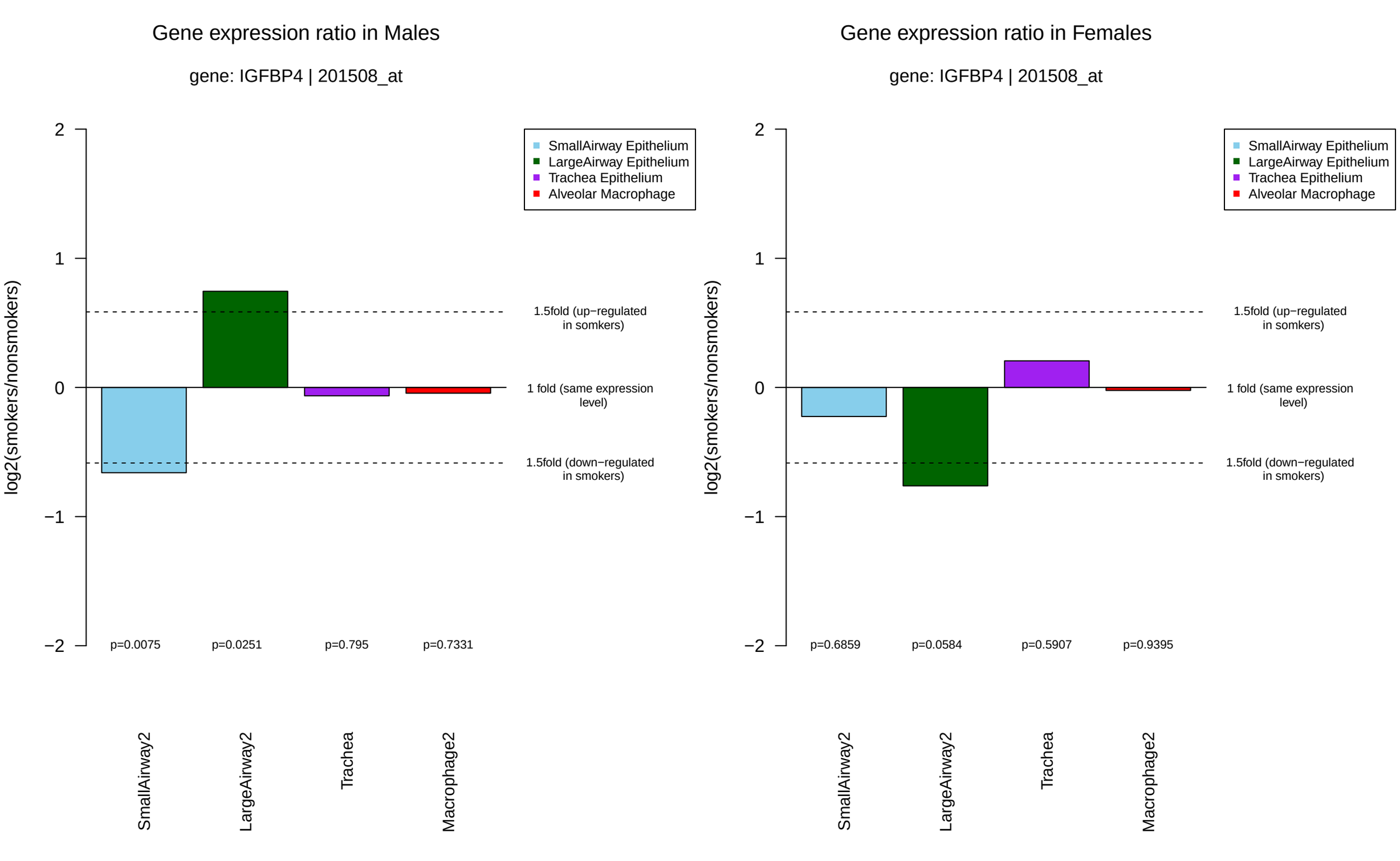
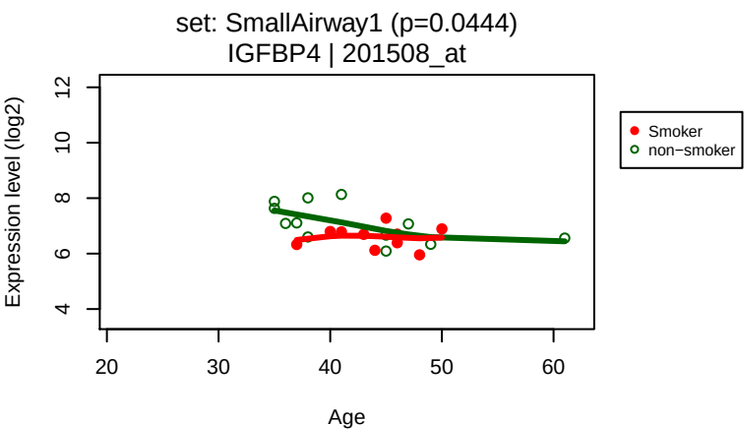
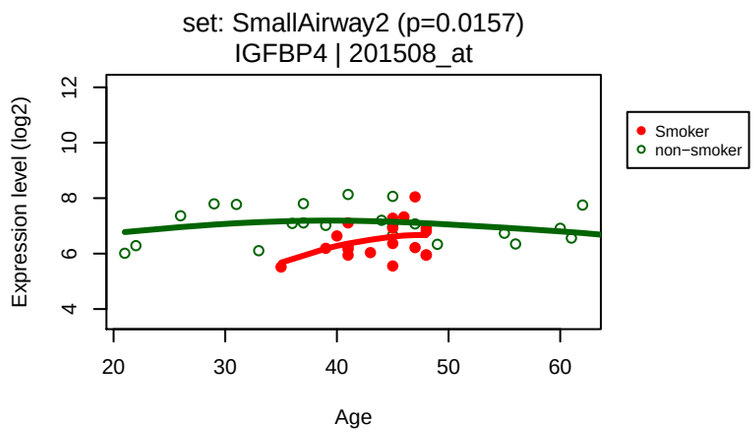
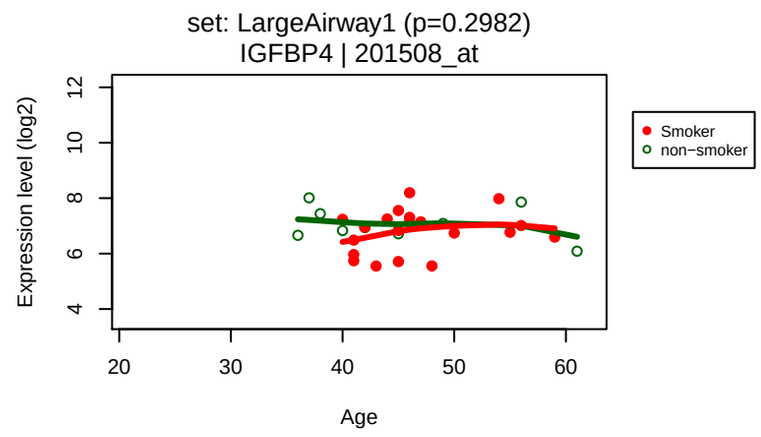

Gene expression of smokers and nonsmokers
gene: IGFBP4 | 201508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0444
p=0.0157
p=0.0123
p=0.2982
p=0.3648
p=0.6424
p=0.7625
p=0.8127
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

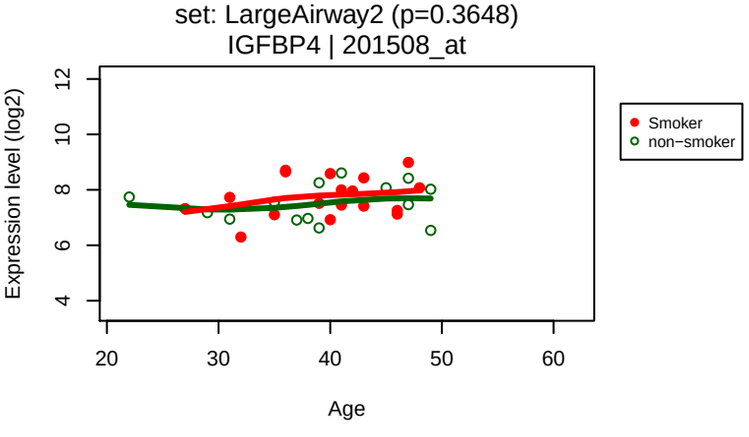

Gene expression ratio (smokers/nonsmokers)
gene: IGFBP4 | 201508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0444
p=0.0157
p=0.0123
p=0.2982
p=0.3648
p=0.6424
p=0.7625
p=0.8127
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

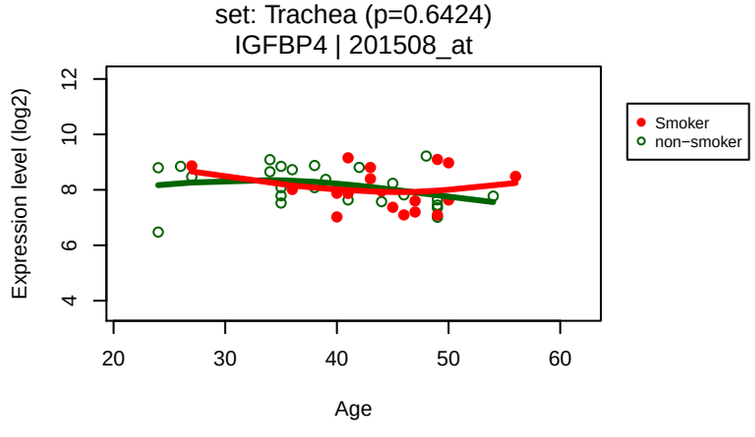

Gene expression ratio in Males
gene: IGFBP4 | 201508_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0075
p=0.0251
p=0.795
p=0.7331
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGFBP4 | 201508_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6859
p=0.0584
p=0.5907
p=0.9395
SmallAirway2
LargeAirway2
Trachea
Macrophage2

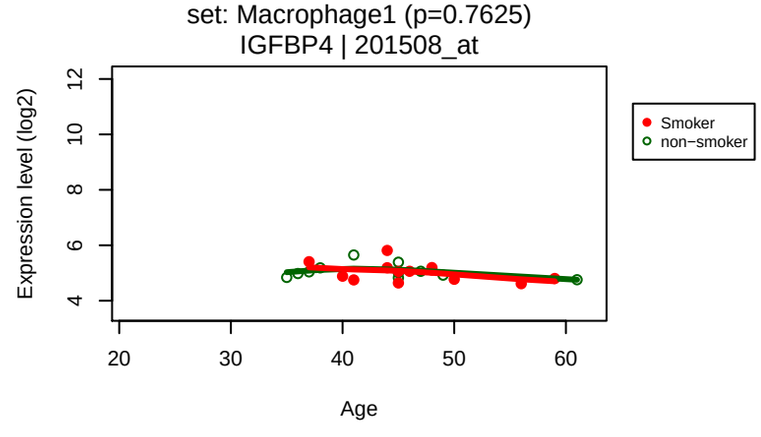

set: SmallAirway1 (p=0.0444)
IGFBP4 | 201508_at
Expression level (log2)
Age
Smoker
non−smoker

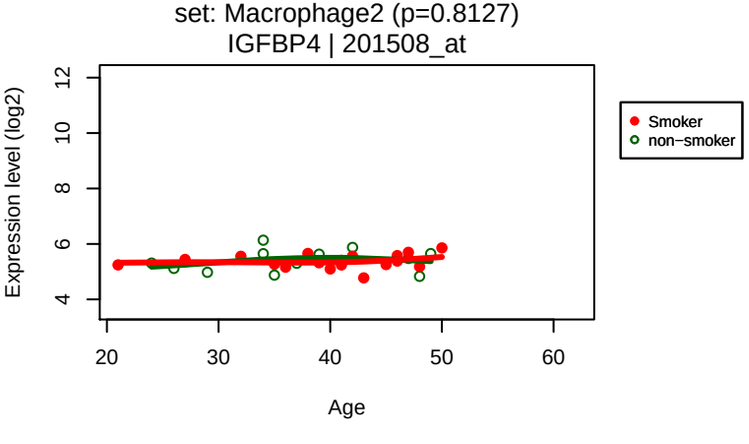

set: SmallAirway2 (p=0.0157)
IGFBP4 | 201508_at
Expression level (log2)
Age
Smoker
non−smoker

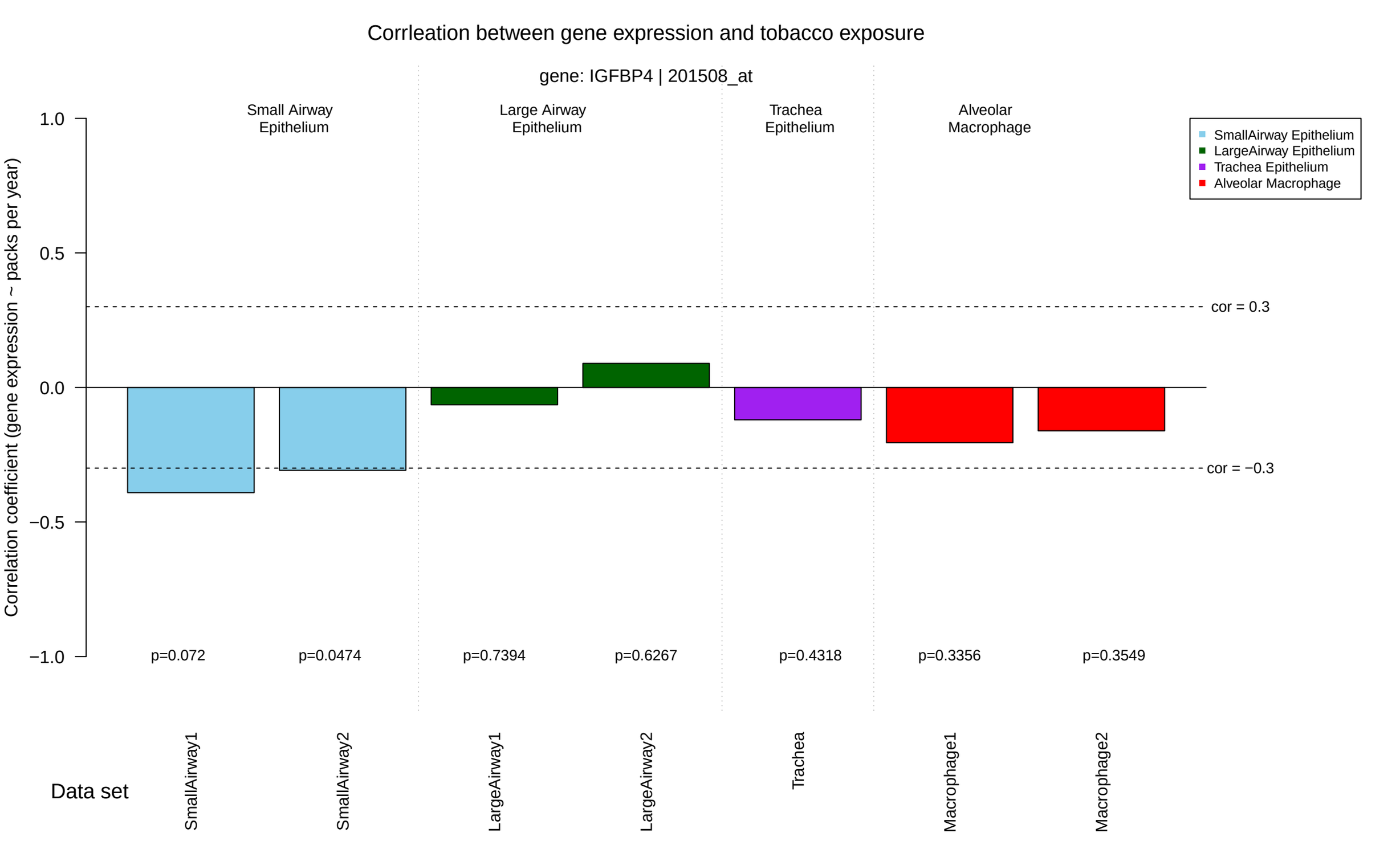

set: LargeAirway1 (p=0.2982)
IGFBP4 | 201508_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3648)
IGFBP4 | 201508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6424)
IGFBP4 | 201508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7625)
IGFBP4 | 201508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8127)
IGFBP4 | 201508_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGFBP4 | 201508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.072
p=0.0474
p=0.7394
p=0.6267
p=0.4318
p=0.3356
p=0.3549
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2